Experiment 9:

Designing an Interactive Healthcare Executive Dashboard in Tableau

1. Aim To design, build, and deploy an end-to-end interactive Executive Healthcare Dashboard in Tableau Desktop using the healthcare_dataset.csv file to analyze key performance indicators (KPIs), patient demographics, medical conditions, and financial distribution.

2. Objectives

- Connect Tableau Desktop to an external CSV data source and prepare data for analysis.
- Formulate calculated fields using Tableau Syntax (e.g., DATEDIFF for Length of Stay).
- Construct essential visual components: KPI Cards, Category Breakdown Charts, and Time-Series Trends.
- Assemble a unified dashboard layout utilizing layout containers, dynamic visual filters, and interactive slicers.

3. Software & Environment Required

- **Software:** Tableau Desktop (or Tableau Public)
- **Dataset:** healthcare_dataset.csv (55,500 records)

4. Dataset Overview & Data Dictionary

- **Total Rows:** 55,500
- **Key Fields:**
    - Name (Text): Patient name
    - Age (Number): Patient age (13 to 89 years)
    - Medical Condition (Text): Diagnosis (Arthritis, Asthma, Cancer, Diabetes, Hypertension, Obesity)
    - Date of Admission / Discharge Date (Date): Hospital stay duration timestamps
    - Admission Type (Text): Elective, Emergency, Urgent
    - Billing Amount (Number): Total financial charge per admission ($)
    - Insurance Provider (Text): Cigna, Medicare, UnitedHealthcare, Blue Cross, Aetna

5. Step-by-Step Procedure

Step 5.1: Data Connection & Cleaning

1. Launch Tableau Desktop and select **Text File** under the **Connect** pane.
2. Select healthcare_dataset.csv to open the Data Source Page.
3. Verify column data types:
    - Set Date of Admission and Discharge Date to **Date**.
    - Set Billing Amount and Age to **Number (decimal / whole)**.
    - Set Medical Condition and Admission Type to **String**.
4. **Data Source Filter:** Click **Add** under *Filters* (top-right) and add a filter on Billing Amount set to **At least 0** (eliminating negative billing entries).

Step 5.2: Create Calculated Fields

1. Right-click in the Data Pane > **Create Calculated Field**.
2. **Length of Stay:**

Code snippet

DATEDIFF('day', [Date of Admission], [Discharge Date])

3. **Total Patients:**

Code snippet

COUNT([Name])

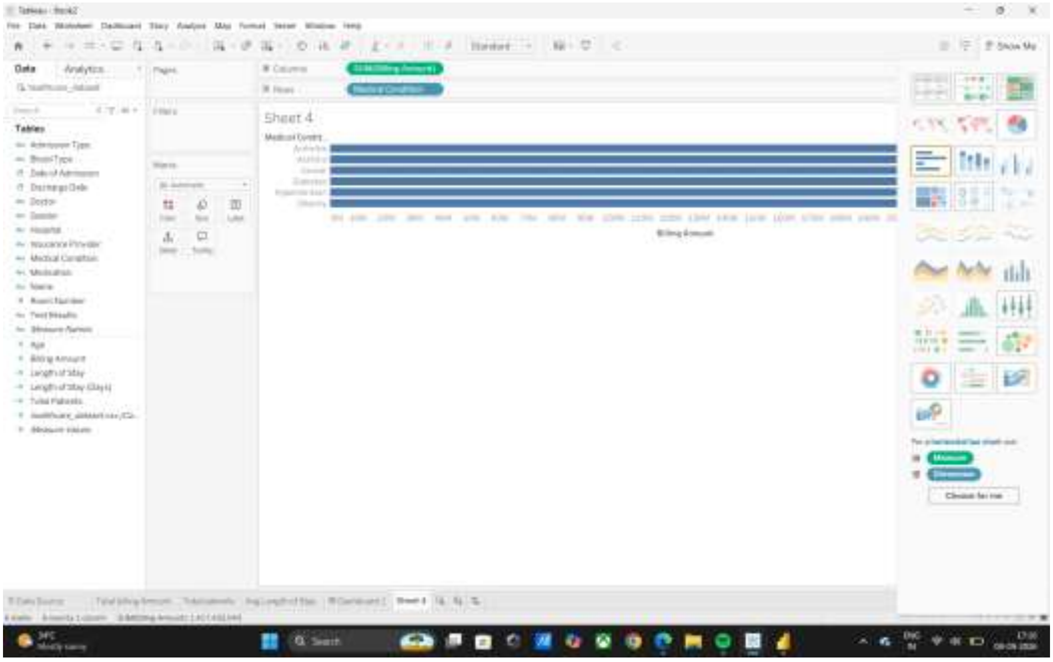
Step 5.3: Build Sheet Visualizations

- **Sheet 1: Total Patients (KPI Card)**
    1. Drag Total Patients to **Text** on the Marks Card.
    2. Format font size to **28pt Bold**. Add text label Total Patients below the value.
    3. Change view from **Standard** to **Entire View**.

- **Sheet 2: Avg Length of Stay (KPI Card)**
    1. Drag Length of Stay to **Text** on the Marks Card.
    2. Change aggregation to **Average**: Right-click pill > **Measure** > **Average**.
    3. Format number format to **1 Decimal Place** (e.g., 15.5 Days). Set to **Entire View**.
- **Sheet 3: Total Billing (KPI Card)**

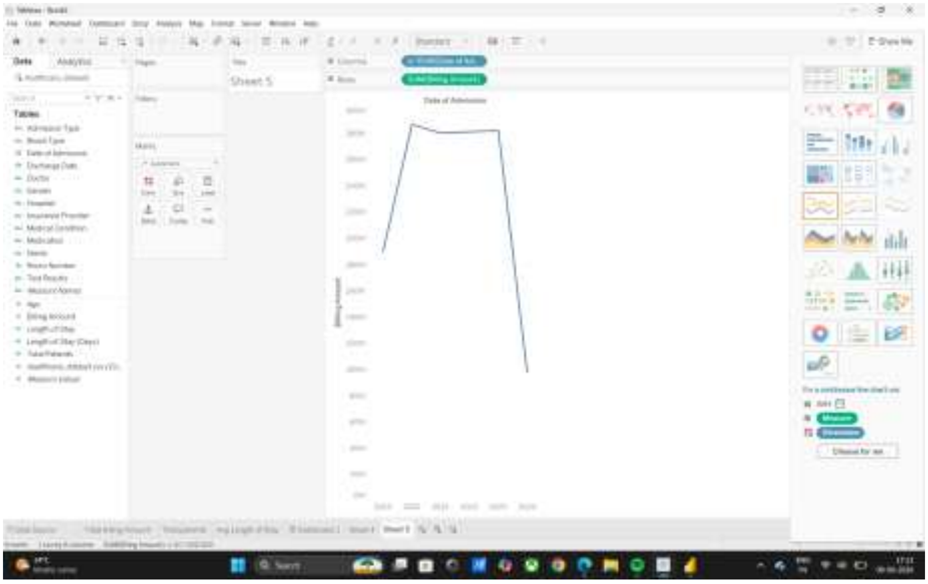
1. Drag Billing Amount to **Text** on the Marks Card.
2. Ensure aggregation is **SUM**. Format number as **Currency (Custom)** in Thousands/Millions.

- **Sheet 4: Billing by Medical Condition (Bar Chart)**
  1. Drag Medical Condition to **Rows**.
  2. Drag Billing Amount (SUM) to **Columns**.
  3. Sort bars in descending order. Drag Medical Condition to **Color** on the Marks Card.



- **Sheet 5: Admission Trends Over Time (Line Chart)**
  1. Drag Date of Admission to **Columns** (Set to **Month/Year**).

2. Drag Total Patients to **Rows**.

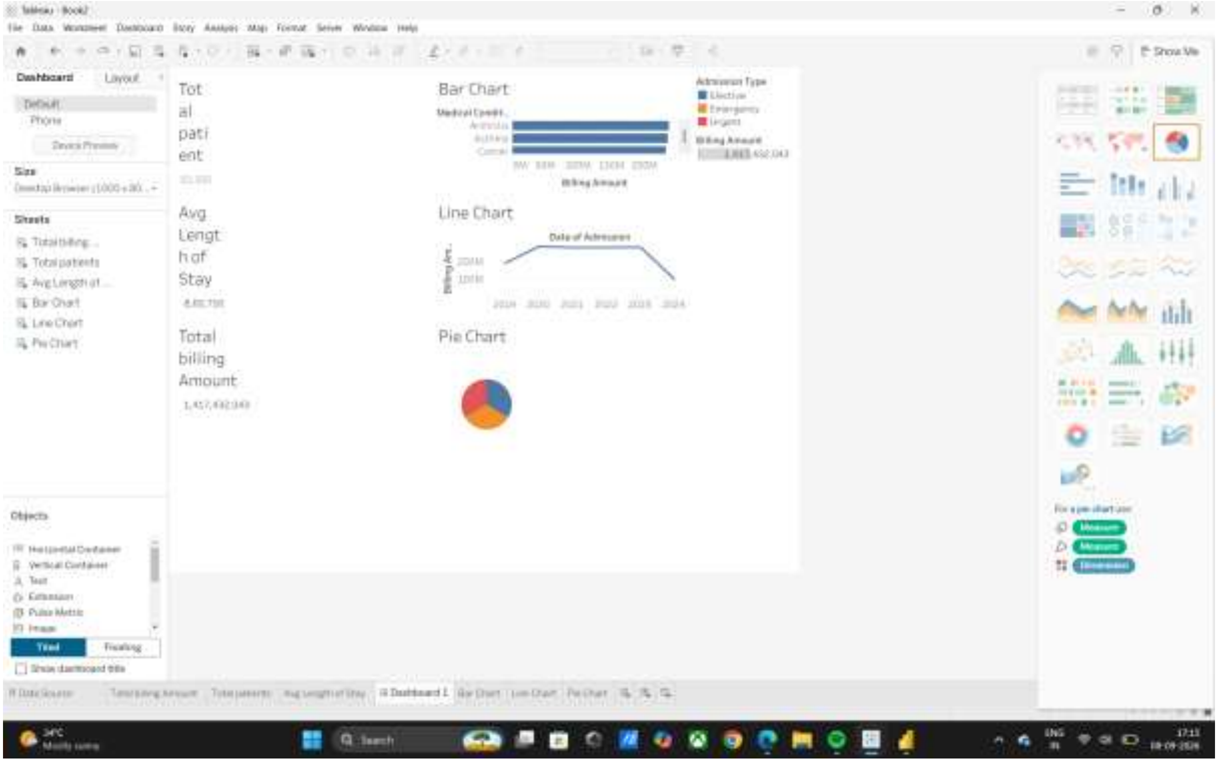


Step 5.4: Dashboard Assembly & Interactivity

1. Click **New Dashboard** tab. Set Size to **Desktop Browser (1366 x 768)**.
2. Drag a **Horizontal Container** to the top section for KPIs.
3. Drop Total Patients, Avg Length of Stay, and Total Billing sheets side-by-side inside the container.
4. Drag Billing by Medical Condition and Admission Trends into the lower canvas area.
5. **Enable Action Filters:** Select the Bar Chart visual, click the **Use as Filter** funnel icon.
6. **Add Interactive Slicer:** Click the menu drop-down on any visual > **Filters** > Admission Type. Set filter scope to **Apply to Worksheets > All Using This Data Source**.

6. Expected Dashboard Output

- **Header / Executive KPI Banner:** Displays top-level metrics ($1.4B Total Billing, 55.5K Total Patients, 15.5 Days Avg Length of Stay).
- **Interactive Filtering:** Selecting a condition (e.g., "Diabetes") dynamically updates all KPI values and time-series line charts across the entire dashboard.



7. **Result :**

   The Healthcare Executive Dashboard was successfully developed in Tableau. The interactive dashboard allows clinical and administrative stakeholders to evaluate patient distribution across medical conditions, monitor financial exposure across insurance providers, and dynamically filter admission trends in real time.